

Social Network Analysis — Unit 1 & Unit 2 Exam Prep

UNIT 1 — Fundamentals of SNA and Network Measures

1. Degree centrality — calculate and identify the most connected node

Concept: Degree centrality of a node = number of edges connected to it (its degree). For a network with n nodes, normalized degree centrality = $\text{degree}(v) / (n - 1)$.

Answer approach:

- Draw/list the adjacency of each node.
 - Count direct connections (degree) for every node.
 - The node with the highest degree count is the "most connected" node — it interacts with the most others directly.
 - Example: If node A connects to B, C, D, E (degree = 4) in a 6-node network, normalized degree centrality = $4/5 = 0.8$.
 - **Interpretation:** High degree centrality = popular/active node, but it only measures *direct* reach, not overall network influence.
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2. Directed & weighted vs undirected & unweighted representation

Answer:

- **Undirected, unweighted:** Represented as a simple adjacency matrix with 0/1 values; a connection A–B implies B–A equally, and all edges are treated as equally strong.
- **Directed, weighted:** Requires an adjacency matrix (or list) where entries indicate direction ($A \rightarrow B \neq B \rightarrow A$) and edge weight (strength/frequency/cost of interaction, e.g., number of messages).
- **Key differences:**
 - Directed graphs distinguish *in-degree* (incoming links) and *out-degree* (outgoing links); undirected graphs have only one "degree."

- Weighted graphs allow centrality/path calculations to factor in strength (e.g., shortest path uses weight sum, not just hop count).
- Directed+weighted networks better model real relationships like "who follows whom" or "how many times X emailed Y," while undirected+unweighted only captures existence of a tie.

3. Betweenness centrality to find a bridge/cut node

Answer:

- **Betweenness centrality** measures how often a node lies on the shortest path between other pairs of nodes: $BC(v) = \sum (\sigma_{st}(v) / \sigma_{st})$, summed over all pairs $s, t \neq v$, where σ_{st} = number of shortest paths from s to t , and $\sigma_{st}(v)$ = number of those passing through v .
 - A node connecting two otherwise disconnected groups will lie on **almost every shortest path** between members of the two groups (since it's the only route), giving it a very high betweenness score even if its degree is low.
 - This node acts as a **"bridge" or "broker"** — removing it would disconnect (or greatly lengthen paths between) the two groups.
 - **Application:** Identifying such nodes is useful for finding gatekeepers, weak points in network resilience, or key influencers who connect communities.
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4. Closeness centrality vs Betweenness centrality

Aspect	Closeness Centrality	Betweenness Centrality
Formula	$CC(v) = (n-1) / \sum d(v,u)$ — inverse of avg. shortest path distance to all others	$BC(v) = \sum \sigma_{st}(v) / \sigma_{st}$ over all pairs
Measures	How quickly a node can reach everyone else	How much a node controls/brokers information flow between others
High value means	Node is centrally located, can spread info fast	Node is a bridge/broker between groups
Application	Closeness: Choosing a node to broadcast an announcement	Betweenness: Identifying a person whose removal would fragment the

scenario	fastest to the whole network (e.g., a PR message)	network (e.g., key connector between departments)
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5. User connected to influential users but not high degree — which centrality?

Answer: Eigenvector centrality (or Katz centrality).

- Eigenvector centrality doesn't just count connections — it weighs each connection by the importance/centrality of the neighbor. A node connected to few but highly influential nodes scores high even with a low raw degree.
- Formula (conceptually): $x_v = (1/\lambda) \sum (A_{vu} \cdot x_u)$, i.e., a node's score is proportional to the sum of its neighbors' scores.
- This models real-world influence: knowing a handful of powerful people can matter more than knowing many unimportant ones.

6. Eigenvector centrality vs Katz centrality

Answer:

- **Eigenvector centrality:** A node's importance is recursively defined by the importance of its neighbors ($x_v \propto \sum \text{neighbors}' x_u$). Works well in strongly connected networks, but fails in directed acyclic graphs or networks with nodes having zero in-links (their score becomes 0, even if they're the source of a chain).
- **Katz centrality:** Adds a constant "base score" (β) to every node, plus an attenuation factor (α) that discounts influence from distant connections: $x_v = \alpha \sum (A_{vu} \cdot x_u) + \beta$
- **Treatment of indirect connections:**
 - Eigenvector centrality only propagates influence through mutually reinforcing (often undirected/strongly connected) structures — indirect paths matter but can vanish to zero in sparse/directed graphs.
 - Katz centrality explicitly counts *all* paths (direct + indirect) to a node, with longer paths contributing exponentially less (via $\alpha^{\text{path length}}$), and works even when a node has no incoming links, because of the β constant.

7. Modeling a student interaction network in NetworkX — steps

Answer (steps):

1. **Data collection:** Gather interaction data (e.g., messages, group projects, class attendance) as an edge list (student A, student B, [weight]).
 2. **Graph construction:** Use `networkx.Graph()` (undirected) or `networkx.DiGraph()` (directed) and add nodes/edges: `G.add_edge(u, v, weight=w)`.
 3. **Basic exploration:** Check `G.number_of_nodes()`, `G.number_of_edges()`, `nx.density(G)`.
 4. **Compute centrality measures:** `nx.degree_centrality(G)`, `nx.betweenness_centrality(G)`, `nx.closeness_centrality(G)`, `nx.eigenvector_centrality(G)`.
 5. **Community/structure analysis:** `nx.connected_components(G)`, clustering coefficient `nx.clustering(G)`, community detection (e.g., `nx.community.greedy_modularity_communities()`).
 6. **Visualization:** `nx.draw(G, with_labels=True)` using spring/circular layouts to inspect structure visually.
 7. **Interpretation:** Identify most influential/isolated students, tightly-knit groups, and potential bridges between friend groups.
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8. Gephi for exploring/visualizing a social media network

Answer:

- Gephi is an open-source network visualization tool that imports network data (CSV/GEXF/GraphML) and renders interactive, force-directed layouts, letting users visually explore structure, apply layout algorithms (ForceAtlas2, Fruchterman-Reingold), filter/color nodes by attribute, and compute statistics.
 - **Two types of information obtainable from visualization:**
 1. **Community structure:** Node clusters/colors (via modularity-based coloring) reveal tightly connected sub-groups or communities within the larger network.
 2. **Influential/central nodes:** Node size (mapped to degree/betweenness/PageRank) visually highlights the most important or highly connected users at a glance, along with bridge nodes connecting different clusters.
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9. Combining centrality measures for influence ranking (professional network)

Answer:

- No single centrality captures "influence" fully, so platforms often combine multiple measures:
 - **Degree centrality** → immediate popularity/reach.
 - **Eigenvector/Katz centrality** → quality of connections (influential neighbors).
 - **Betweenness centrality** → brokerage/bridge role across communities.
 - **Closeness centrality** → speed of spreading information.
 - **Combination approach:** Normalize each measure (0–1 scale) and compute a weighted composite score: $\text{Influence Score} = w_1 \cdot \text{Degree} + w_2 \cdot \text{Eigenvector} + w_3 \cdot \text{Betweenness} + w_4 \cdot \text{Closeness}$ where weights ($w_1..w_4$) are tuned based on business goals (e.g., emphasize eigenvector for "thought leaders," betweenness for "connectors").
 - This multi-metric approach avoids bias of any single measure (e.g., a node could have high degree but low real influence, or vice versa).
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10. Bipartite network for users + companies

Answer:

- **Yes, a bipartite representation is appropriate.** A bipartite graph has two disjoint node sets (Users, Companies) where edges only exist *between* the sets, never within a set (i.e., no user–user or company–company edges directly).
 - **Justification:**
 - The actual relationships recorded are "user interacts/follows/works-for company," which is naturally a two-mode relationship.
 - Bipartite modeling avoids forcing an artificial single-mode structure and preserves the true nature of interactions.
 - It also enables useful derived analyses: **projection** onto a one-mode "user–user" network (users sharing common companies) or "company–company" network (companies sharing common users) for further centrality/community analysis.
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UNIT 2 — Network Growth Models and Link Analysis

11. Erdos-Renyi vs Small-World vs Scale-Free — structural comparison

Model	Degree Distribution	Clustering	Avg. Path Length	Real-world fit
Erdos-Renyi (Random)	Poisson (binomial) — most nodes have similar degree	Low	Short	Poor fit — real networks aren't purely random
Small-World (Watts-Strogatz)	Fairly uniform, but with some rewired long-range links	High (like regular lattices)	Short (like random graphs)	Good for networks with local clustering + short global paths (e.g., friendship networks)
Scale-Free (Barabási-Albert)	Power-law — few hub nodes, many low-degree nodes	Moderate	Short	Fits many social/web networks with hubs (celebrities, popular pages)

12. Preferential attachment for a growing social network

Answer:

- The **preferential attachment model** (Barabási–Albert) states: new nodes joining the network are more likely to connect to nodes that already have high degree ("rich get richer").
- **Application:** As new users join a growing social platform, they're statistically more likely to follow/friend already-popular users (celebrities, early adopters) rather than random unknown users.
- Mathematically, probability of connecting to node i : $P(i) = k_i / \sum_j k_j$, where k_i is the degree of node i .
- Over time, this creates a small number of very high-degree "hub" nodes and a long tail of low-degree nodes — producing a **power-law (scale-free) degree distribution**.

13. How preferential attachment creates highly connected (hub) nodes

Answer:

- Early nodes have a "first-mover advantage": since new nodes preferentially attach to already-popular nodes, nodes that gained a few extra links early accumulate disproportionately more links over time.
 - This is a **positive feedback loop** ("rich get richer" / Matthew effect): higher degree → higher probability of new connections → even higher degree.
 - Result: a small number of nodes become hubs with very high degree, while most nodes remain low-degree — producing the heavy-tailed power-law distribution characteristic of scale-free networks.
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14. High clustering + short paths → which model?

Answer: Small-World model (Watts-Strogatz).

- **Justification:**
 - High clustering coefficient means nodes' neighbors are also connected to each other (tightly-knit local groups) — characteristic of regular/lattice-like structure.
 - Short average path length between *any* two nodes (not just neighbors) is NOT typical of regular lattices (which have long paths) — it comes from a few random "shortcut" edges rewired across the network.
 - The Watts-Strogatz model starts with a regular ring lattice (high clustering) and randomly rewires a small fraction of edges (creating shortcuts, hence short paths) — exactly producing this combination, known as the "small-world phenomenon" (e.g., "six degrees of separation").
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15. Random growth vs preferential attachment — emergence of highly connected nodes

Answer:

- **Random network growth (Erdos-Renyi style):** Each new edge/node connects to existing nodes with **equal probability**, regardless of current degree. This produces a fairly homogeneous, Poisson-distributed degree — no single node becomes dramatically more connected than others; hubs essentially don't emerge.

- **Preferential attachment:** New nodes connect with probability proportional to existing degree, so already well-connected nodes keep gaining more links faster than others.
 - **Outcome:** Random growth → uniform, "egalitarian" network with no real hubs.
Preferential attachment → skewed, power-law network with a few dominant hubs and many low-degree nodes — matching real social/web network structures far better.
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16. PageRank for ranking web pages

Answer:

- **PageRank** models a page's importance based on the number and quality of pages linking to it — a link from an important page counts more than a link from an unimportant one.
 - Formula: $PR(A) = (1-d)/N + d \cdot \sum (PR(T_i)/C(T_i))$, where T_i are pages linking to A, $C(T_i)$ is the out-degree of T_i , d is the damping factor (~ 0.85), N is total pages.
 - **Application to search ranking:**
 - Model the web as a directed graph (pages = nodes, hyperlinks = directed edges).
 - Compute PageRank iteratively (or via eigenvector of the link matrix) until scores converge.
 - Pages with higher PageRank are considered more authoritative/important and are ranked higher in search results, combined with keyword relevance.
 - The damping factor models a "random surfer" who occasionally jumps to a random page instead of following links, preventing rank sinks in disconnected/looped structures.
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17. HITS — Authority and Hub scores

Answer:

- **HITS (Hyperlink-Induced Topic Search)** assigns two scores to each page:
 - **Authority score:** how valuable a page's *content* is, based on how many good **hubs** link to it.
 - **Hub score:** how good a page is as a *directory/index*, based on how many good **authorities** it links to.

- **Mutual reinforcement (recursive definition):**
 - $\text{Authority}(p) = \sum \text{Hub}(q)$ for all q linking to p
 - $\text{Hub}(p) = \sum \text{Authority}(r)$ for all r that p links to
- **Example:** Imagine a "Best AI Blogs" listing page (hub) that links to several high-quality AI research pages (authorities). The listing page's hub score rises because it links to good authorities; those research pages' authority scores rise because a good hub links to them. This is computed iteratively (like power iteration) until scores converge.

18. PageRank vs HITS

Aspect	PageRank	HITS
Scope	Computed globally, once, for the entire web graph (query-independent)	Computed dynamically on a small subgraph relevant to a specific query (query-dependent)
Scores	Single importance score per page	Two scores per page: Hub and Authority
Basis	Random-surfer model; link "votes" propagated with damping	Mutual reinforcement between hubs and authorities
Use case	General-purpose ranking (e.g., Google's original ranking)	Topic-specific ranking (e.g., finding good resource lists vs. authoritative content for a query)

19. Snapshot networks and temporal links for evolving social networks

Answer:

- **Snapshot networks:** Capture the network's state at discrete time points (e.g., network as of Day 1, Day 2, Day 3), essentially creating a sequence of static graphs. Comparing snapshots reveals growth, shrinkage, or structural shifts (e.g., new communities forming) over time.
- **Temporal links:** Instead of treating edges as permanently present, each edge is tagged with a timestamp (or time interval) indicating exactly when the interaction occurred, enabling analysis of the *order* and *timing* of connections (not just their existence).
- **Combined use:**

- Track evolving centrality (e.g., is a node's influence growing?).
 - Detect emerging/dissolving communities over time.
 - Study information/diffusion cascades respecting actual chronological order of edges (a temporal path is only valid if edge timestamps are non-decreasing along the path).
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20. Personalized PageRank for user-specific recommendations

Answer:

- Standard PageRank produces one global importance score per node, ignoring individual user preferences.
 - **Personalized PageRank (PPR)** modifies the "random jump" component: instead of jumping uniformly to any random page, the surfer jumps back to a **specific set of seed nodes** representing a user's interests/past interactions (e.g., videos they watched, topics they follow).
 - Formula: $PR(A) = (1-d) \cdot e_u(A) + d \cdot \sum (PR(T_i)/C(T_i))$, where e_u is a personalization vector biased toward the user's preferred nodes instead of uniform $1/N$.
 - **Application:** A video platform builds one personalization vector per user (or per user cluster) based on their watch history, then computes PPR — nodes/videos "closer" (in link/similarity structure) to the user's seed interests rank higher, producing individualized recommendations even from the same underlying content graph.
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Extra Practice Questions (Unit 1 & 2)

1. Define network density. How does it differ between a sparse and a dense social network?
2. What is the clustering coefficient of a node, and what does a high value indicate about a user's friend circle?
3. Explain the difference between a walk, a path, and a cycle in graph theory with examples.
4. Two nodes have the same degree but different betweenness centrality. Explain how this is possible and what it implies.
5. How would you represent a multigraph (multiple types of relationships between the

same two users, e.g., "friend" and "colleague")? Give a use case.

6. Compare adjacency matrix and adjacency list representations of a graph in terms of memory efficiency for sparse vs. dense networks.
 7. What is a "structural hole" in a social network, and how does it relate to betweenness centrality?
 8. Explain how the Barabási–Albert model differs from the Watts–Strogatz model in terms of the network-generation mechanism (growth vs. rewiring).
 9. A network shows a power-law degree distribution. What does this imply about the network's robustness to random node failure vs. targeted attack on hubs?
 10. Explain the concept of "six degrees of separation" and how it relates to the small-world property.
 11. What is the difference between local and global clustering coefficient?
 12. Given a directed graph, explain how PageRank handles "dangling nodes" (pages with no outgoing links).
 13. Compare weighted PageRank and standard PageRank — how does incorporating edge weights change the ranking?
 14. Explain how community detection (e.g., modularity-based) differs fundamentally from centrality-based analysis — what question does each answer?
 15. A recommendation system wants to suggest new connections ("people you may know"). Explain how link prediction techniques (e.g., common neighbors, Jaccard coefficient) could be applied.
 16. Explain why eigenvector centrality can give misleading (zero) scores in a directed acyclic graph, and how Katz centrality resolves this.
 17. What is the difference between an "ego network" and the full social network? Give a scenario where ego-network analysis is more useful.
 18. Explain assortativity/degree correlation in a network — what does it mean if high-degree nodes tend to connect to other high-degree nodes?
 19. How does the choice of damping factor (d) in PageRank affect the ranking results? What happens at $d = 0$ and $d = 1$?
 20. Explain how the "rich-get-richer" phenomenon in preferential attachment relates to real-world inequality in follower counts on social media platforms.
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Tip for exam: For "explain/compare" questions, always structure your answer as **Definition → Formula/Mechanism → Example → Real-world Application**. For "analyze/identify" scenario questions, name the correct measure/model first, then justify with 2–3 reasoning points.